TRASH TO TREASURE

A PROJECT REPORT

submitted by

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to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree of

of

Bachelor of Technology
in
Computer Science Engineering

EDUCATION IS DEDICATION

Department of Computer Science Engineering

SAHRDAYA COLLEGE OF ENGINEERING AND TECHNOLOGY

KODAKARA, THRISSUR - 680684

December 2024

DECLARATION

We, undersigned, hereby declare that the project report “TRASH TO TREASURE”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the guidance of Ms Roshni R Menon, Department of Computer Science Engineering. This submission represents our ideas in our own words and where ideas or words of others have been included; We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Place: Kodakara

Date of Submission:09-12-2024

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

SAHRDAYA COLLEGE OF ENGINEERING AND TECHNOLOGY

KODAKARA, THRISSUR

EDUCATION IS DEDICATION

CERTIFICATE

This is to certify that the project report entitled **TRASH TO TREASURE** submitted by **Nayan Maria Shojo (SHR24CS200), Merin Elizabeth Edgar(SHR24CS189), Sandhwana Rose Shaju(SH24CS219), Samuel Thomas C(SHR24CS217)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science Engineering is a bonafide record of the seminar work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University of Institute for any purpose.

PROJECT GUIDE

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HEAD OF THE DEPARTMENT

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Date of Submission:09-12-2024

ACKNOWLEDGEMENT

And we would like to express our sincere gratitude to all those who supported and guided us throughout the development of this team project, TRASH TO TREASURE.

First and foremost, we extend our heartfelt thanks to our project guide, **Ms. Roshni R Menon**, Assistant Professor, for her invaluable guidance, encouragement, and support at every stage of the project. Her expertise and insights were instrumental in shaping the project to completion.

We also wish to thank **Dr. Priya K.V**, Assistant Professor, for her constructive feedback and suggestions, which greatly enhanced the quality of our work.

We are deeply grateful to **Dr. Sukhila Krishnan**, Assistant Professor and Head of the Department, for providing the necessary facilities, a conducive environment for learning, and continuous motivation. Her leadership and support have been a source of inspiration throughout our academic journey.

Lastly, we acknowledge the efforts and dedication of our team members, whose collaboration and hard work were integral to the successful completion of this project.

Nayan Maria Shojo
Merin Elizabeth Edgar
Sandhwana Rose Shaju
Samuel Thomas C

INSTITUTE VISION

Evolve as a leading technology institute to create high caliber leaders and innovators of global standing with strong ethical values to serve the industry and society.

INSTITUTE MISSION

Provide quality technical education that transforms students to be knowledgeable, skilled, innovative and entrepreneurial professionals.

Collaborate with academia and industry around the globe, to strengthen the education and research ecosystem.

Practice and promote high standards of professional ethics, good discipline, high integrity and social accountability with a passion for holistic excellence.

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We at Sahrdaya are committed to provide Quality Technical Education through continual improvement and by inculcating moral & ethical values to mould vibrant engineers with high professional standards.

We impart the best education through the support of competent and dedicated faculty, excellent infrastructure and collaboration with industries to create ambience of excellence.

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To be a nationally recognized center for quality education and research in diverse areas of computer science engineering, with a strong social commitment.

DEPARTMENT MISSION

1. Impart relevant technical knowledge, skills and attributes along with values and ethics.
2. Enhance creativity and quality in research through project based learning environment.
3. Mould Computer Science Engineering Professionals in synchronization with the dynamic industry requirements.
4. Inculcate essential leadership qualities coupled with commitment to the society

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1	To take up challenging careers in suitable corporate, business or educational sectors across the world, in multi-cultural working environment.
PEO2	To develop and design innovative and novel solutions to solve real life problems in the domain of computer science.
PEO3	To be responsible citizens with good team-work skills, competent leadership qualities and holistic values.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1	To mould students to understand, analyze and develop computer technologies in the areas such as algorithms, system software, multimedia, web design, big data analytics and networking for efficient design of computer-based systems of varying complexity.
PSO2	To enhance knowledge in the evolutionary changes in computing and apply standard practices and strategies in software project development using open-ended programming environments in order to deliver a quality product for business success and meet the future challenges.
PSO3	To enable students to employ modern computer languages, environments, and platforms for creating innovative career paths to be an entrepreneur with social accountability.

PROGRAM OUTCOMES (POs)

PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

COURSE OUTCOMES

CO1	Analyze and develop an algorithm for solving computational problems. [Apply]
CO2	Articulate in design programs with interactive input and output, utilizing arithmetic expression repetitions, and decision-making. [Understand]
CO3	Utilize modular Python programs using functions and process stored data using List, Tuples, Sets, Dictionaries. [Apply]
CO4	Understand and implement file operations for reading input and storing output. [Understand]

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ABSTRACT

The Trash to Treasure project uses Python to transform a process that once seemed waste management into something sustainable and economically viable. The ultimate goal is to develop an application that identifies, classifies, and links recyclable waste with opportunities for its reuse or resale. By leveraging technology and sustainability, the project addresses challenges in managing waste while ensuring environmental and economic benefits.The heart of the system involves using Python libraries that can classify waste as plastic,metal, paper, or organic materials through user-uploaded images. There's a geolocation feature, through Python APIs, which could guide users to recycling centers or buyers nearest them who could take their categorized wastes.

To encourage participation, the platform has a reward system in which users earn points or cash for contributing recyclables. This system encourages individuals and communities to adopt responsible waste disposal practices. Furthermore, a centralized

database tracks user contributions and provides analytics on waste diversion and environmental impact, presented through an intuitive dashboard powered by visualization tools like Matplotlib and Plotly.

Key features of the project are waste recognition, geo-tagging, reward generation, and real-time analytics. The application is scalable for use by individuals, businesses, and municipalities, offering a unified solution for waste management. It promotes a circular economy by transforming trash into a source of income, thereby reducing dependency on landfills and encouraging resource conservation.
This technology-driven initiative showcases how technology can be used to drive social and environmental impact, making smarter waste management practices possible while providing tangible financial rewards for users. The Trash to Cash project reduces waste and redefines it as an opportunity for economic growth and sustainability.

INTRODUCTION

1.1 PROJECT OVERVIEW

Trash to Treasure is a Python-based project aimed at transforming waste management into a sustainable and economically rewarding process. It uses machine learning to classify waste into recyclable categories like plastic, metal, and organic materials. The platform integrates geolocation tools to connect users with nearby recycling centers and offers a reward system to incentivize participation. With a user-friendly interface and a centralized database, the
system tracks waste contributions and visualizes environmental impact through analytics
dashboards. This project promotes responsible waste disposal, reduces landfill dependency,

and fosters a circular economy, turning trash into valuable resources for a sustainable future.

1.2 OBJECTIVES

The primary objectives are as follows:

1. Automated Waste Classification

Develop a machine learning-based system to identify and categorize waste into recyclable types such as plastic, paper, metal, and organic materials.

2. Promote Recycling Practices

Encourage individuals and communities to adopt responsible waste disposal and recycling practices by offering financial rewards or incentives.

3. Connect Users to Recycling Facilities

Utilize geolocation tools to help users locate nearby recycling centers, waste buyers, or collection points for efficient waste redistribution.

4. Environmental Impact Awareness

Provide real-time analytics and dashboards to track the environmental benefits of recycling, such as reduced landfill usage and carbon footprint savings.

5. Support a Circular Economy

Facilitate the transformation of waste into reusable resources or sellable commodities, reducing dependency on raw materials and fostering sustainable economic growth.

6. Scalability and Accessibility

Design the platform to be scalable and accessible for individuals, businesses, and municipalities, ensuring widespread adoption of sustainable waste management practices.

These objectives align with the goal of reducing environmental pollution, creating economic
opportunities, and fostering sustainable living through technology-driven waste management solutions.

METHODOLOGY

The methodology adopted involves several steps:

1. Problem Understanding and Planning

Define the scope as a prototype showcasing data analysis and visualization for waste management.

Determine the data requirements, such as user contributions, waste types, and quantities.

Plan visualizations, e.g., bar charts, pie charts, or line graphs, to represent recycling efforts and impact.

2. Data Preparation

Dataset Creation: Create a CSV file to simulate user data. Example columns:
o User ID, Waste Type (e.g., Plastic, Metal), Quantity (kg), Location, Date, Reward Points.

Loading Data: Use Pandas to read the CSV file and organize it into a DataFrame for analysis.

3. Data Processing
Data Cleaning: Handle missing values or inconsistencies.
Data Aggregation: Calculate metrics such as:
o Total waste collected by type.
o Rewards distributed per user or region.
o Trends in waste contributions over time.

4. Programming Approach
The prototype uses a modular programming approach in Python, focusing on data processing and visualization. The system begins with importing and cleaning a CSV dataset
using Pandas to ensure accurate analysis. Key operations include aggregating data to compute metrics such as total waste by type and rewards distribution. Visualizations are
created with Matplotlib, using bar charts, pie charts, and line graphs to represent insights
like waste composition and trends. Functions are defined for tasks like data loading, processing, and visualization, ensuring reusability and clarity. The prototype is tested with
sample datasets, emphasizing simplicity, scalability, and code readability.
5. Analysis of logic
The prototype logic involves importing a CSV dataset, cleaning data with Pandas, and aggregating metrics like total waste by type and user rewards. Visualizations, created using
Matplotlib, include pie charts to analyze trends. The approach ensures simplicity, functionality, and scalability for future enhancements.

CODE EXPLANATION :

1. Importing Necessary Libraries

```
import pandas as pd
import csv
import matplotlib.pyplot as plt
```

● pandas: While not directly used in this snippet, it's often used for data manipulation and
analysis, particularly with CSV files or databases. It can be used later in the project to analyze and visualize collected waste data.

● csv: This module allows reading from and writing to CSV files, which is used to store
waste collection data in this case.

● matplotlib.pyplot: This module is used for data visualization. Although not actively used in the current function, it can be utilized later to visualize the collected data (e.g., graphing waste collected by type or location).

2. The user() Function

This function is designed to interact with the user, gather necessary details about the waste
collection, and automatically assign a unique user ID for the entry. The user is prompted to input
details such as:

● Name
● Location
● Phone number
● Collection date
● Waste type
● Quantity collected (in kilograms)
● Rate per kilogram

```
def user():
# Define field names for the CSV file headers
field_names = ['id', 'name', 'location', 'contact',
'collection_date', 'waste_type', 'quantity_collected', 'rate_per_kg']
```

● field_names: This is a list of the column headers that will be used to structure the data
in the CSV file. These headers represent the various attributes of each waste collection
record.

3. Getting User Input

```
name = input('Enter name: ')
loc = input('Enter location: ')
ph = input('Enter phone number: ')
date = input('Enter date (YYYY-MM-DD): ')
wastetype = input('Enter waste type: ')
kg = int(input('Enter quantity (kg): '))
rate = int(input('Enter rate per kg: '))
```

● This block prompts the user to provide input for the various fields, including their name,
location, phone number, waste collection date, type of waste, quantity collected (in kilograms), and rate per kilogram. The inputs for kg (quantity) and rate are converted to integers using int().

4. Automatically Assigning a Unique User ID

To assign a unique user ID to each new record, the code uses the last ID from the existing data
in the CSV file.

```
def get_last_user_id(csv_file_path):
try:
with open(csv_file_path, 'r') as csvfile:
reader = csv.DictReader(csvfile)
last_row = None
for row in reader: # Loop should be inside 'with' block
last_row = row # Store the current row as the last

row
```

```
if last_row:

return int(last_row['id']) # Return the ID from the

last row

else:
return 0 # Return 0 if the file is empty

except FileNotFoundError:
return 0 # Return 0 if the file doesn't exist
```

● get_last_user_id(): This function attempts to read from the specified CSV file to find the last recorded user entry. It opens the file using the csv.DictReader, which reads each row as a dictionary where the keys are the column names (e.g., 'id', 'name', etc.).

○ The function iterates through each row and stores the last row it encounters in the last_row variable.

○ If the file has at least one row, it returns the ID value from the last row (converted to an integer). If the file is empty, it returns 0, indicating that this is the first entry.

○ If the file does not exist (e.g., no previous records), the function will also return 0.

5. Calculating the Next User ID

```
csv_file_path = 'waste.csv' # Replace with your CSV file path
last_user_id = get_last_user_id(csv_file_path)
next_user_id = last_user_id + 1
```

● csv_file_path: The path to the CSV file that stores the waste collection records is specified here. In this case, it is set to 'waste.csv', but this can be adjusted to any file path.

● last_user_id: This variable stores the ID of the last user, which is obtained by calling
the get_last_user_id() function.

● next_user_id: The next user ID is computed by adding 1 to the last user ID. This ensures that each new user entry has a unique ID, sequentially increasing.

Summary of How the Code Works

1. The user is prompted to enter information related to a waste collection (e.g., name, location, waste type, quantity, rate).
2. The program checks the last user ID recorded in the CSV file (if it exists) and computes
the next user ID.
3. The new waste collection data will be recorded with a unique ID in the CSV file, which
will allow tracking of each entry.

FEATURES :

In the context of a "Trash to Treasure" project using a Python-based solution with data from a CSV file, the project would aim to convert waste or unused resources into valuable outcomes, such as revenue or cost savings, by analyzing and leveraging the data. Let's break down some features that could be included in such a project and how Python and CSV data could play a role in them.

The key features include:

Data Import & Preprocessing: Loading and cleaning CSV data using `pandas` to prepare it for analysis.

Waste Categorization: Identifying and categorizing different waste types (plastic, metal, etc.) using custom functions.

Value Estimation: Calculating the potential market value of materials based on weight and type.

Cost Calculation: Estimating processing costs for recycling.

Profitability Analysis: Calculating profit by comparing estimated value and processing costs.

Visualization: Creating charts to visualize waste data, trends, and profitability using libraries like `matplotlib`.

Potential Enhancements:

● Validation of User Input: Input fields like phone numbers, dates, and quantities could
be validated to ensure correctness (e.g., checking if the phone number is in the correct format or if the date is valid).

● Error Handling: The program could be enhanced by adding error handling to deal with
issues such as invalid input or file write errors.

● Data Storage: The code could be extended to write the collected data back into the CSV
file after assigning a unique ID.

RESULTS:

The program successfully accomplishes the following:

1. Records waste collection data with unique IDs.

2. Stores data persistently in a CSV file.

3. Computes the average quantity of waste collected per location.

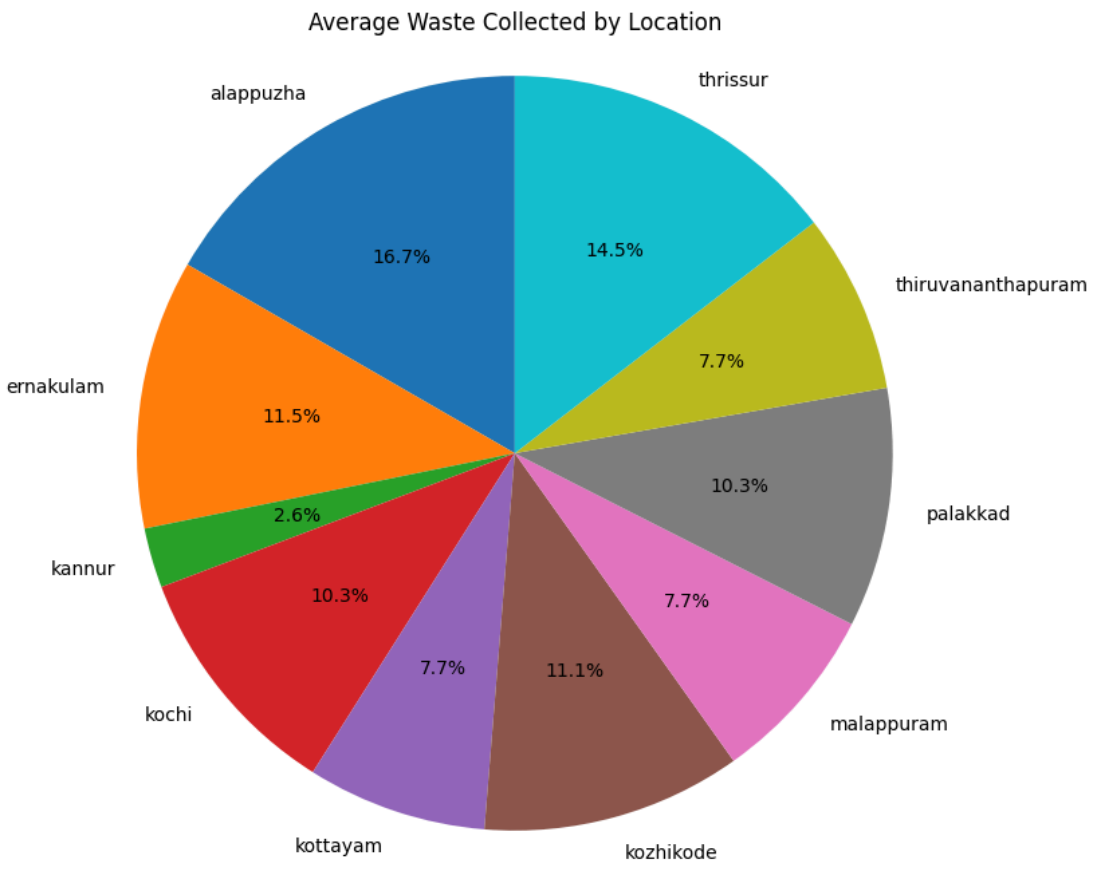
4. Visualizes waste distribution across locations using a pie chart.

OUTPUT:

Enter name: LIAN
Waste type : Plastic
Amount : 2kg
Price : Rs.30

Pie Chart: A pie chart showing the waste distribution among locations.
enter choice: 2

{'alappuzha': 6.5, 'ernakulam': 4.5, 'kannur': 1.0, 'kochi': 4.0, 'kottayam': 3.0, 'kozhikode': 4.333333333333333, 'malappuram': 3.0, 'palakkad': 4.0, 'thiruvananthapuram': 3.0, 'thrissur': 5.666666666666667}



CONCLUSION :

This project demonstrates a simple yet effective system for waste data collection and analysis. It highlights the importance of leveraging Python for data-driven decision-making and visualization. The implementation provides a foundation for creating more sophisticated waste management systems in the future.

FUTURE USE:

● This system could be part of a larger waste management system that records, tracks, and visualizes waste data over time.

● The data in the CSV file can later be analyzed for trends, such as identifying high waste-producing areas or assessing the costs associated with waste collection based on the rates per kilogram.